# Chemistry: CYL1010

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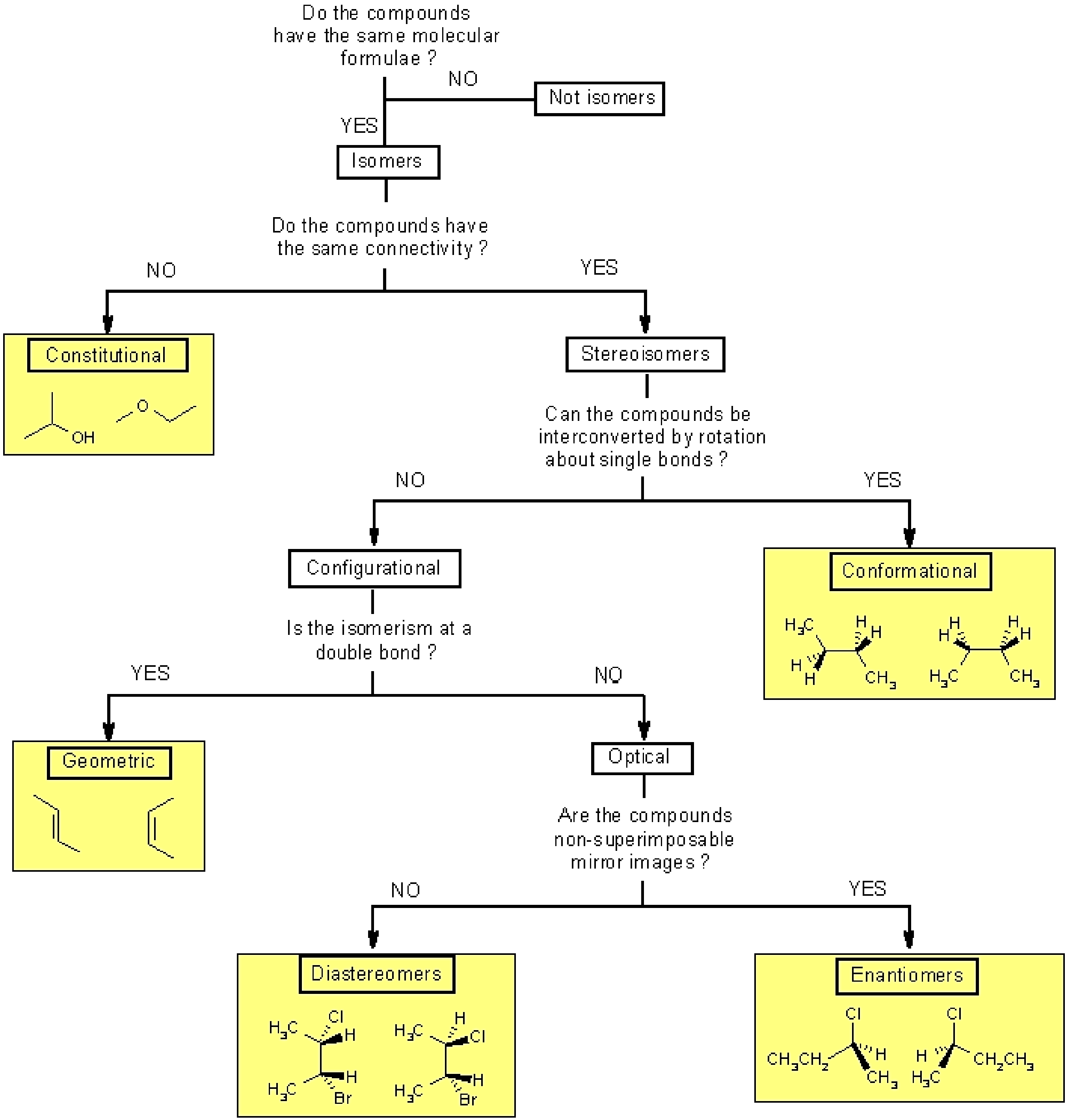
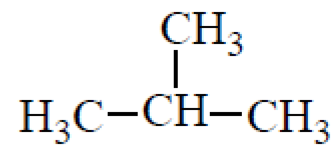
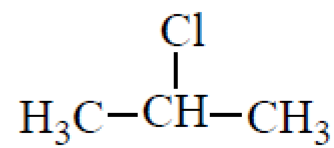
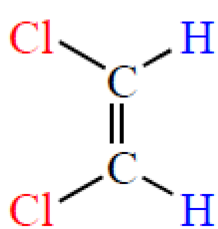
**Indian Institute of Technology Jodhpur**

References:
Organic Chemistry
Textbook by Jonathan Clayden, Nick Greeves, and Stuart Warren

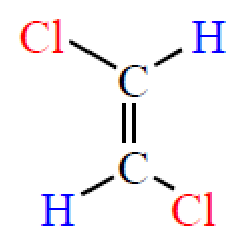
Fundamentals of Organic Chemistry
Book by T. W. Graham Solomons


Do the compounds have the same molecular formulae ?
NO
Not isomers
YES
Isomers
Do the compounds have the same connectivity ?
NO
YES
Constitutional
OH
O
Stereoisomers
Can the compounds be interconverted by rotation about single bonds ?
NO
YES
Configurational
Conformational
Is the isomerism at a double bond ?
YES
NO
Geometric
Optical
Are the compounds non-superimposable mirror images ?
NO
YES
Diastereomers
Enantiomers

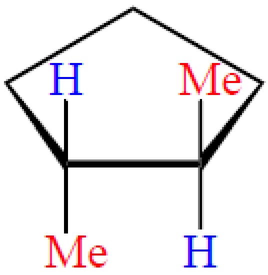
| | | | |
|---|---|---|---|
| $C_4H_{10}$ | $CH_3CH_2CH_2CH_3$<br>**Butane** | and | <br>**Isobutane** |
| $C_3H_7Cl$ | $CH_3CH_2CH_2Cl$<br>**1-Chloropropane** | and | <br>**2-Chloropropane** |
| $C_2H_6O$ | $CH_3CH_2OH$<br>**Ethanol** | and | $CH_3OCH_3$<br>**Dimethyl ether** |



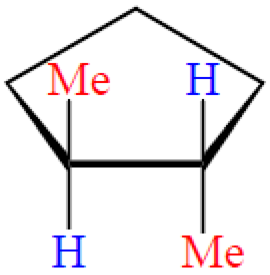
*cis*-1,2-Dichloroethene ($C_2H_2Cl_2$)



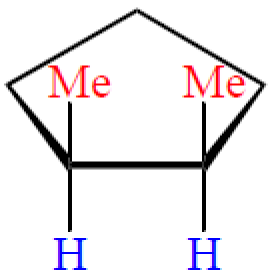
*trans*-1,2-Dichloroethene ($C_2H_2Cl_2$)



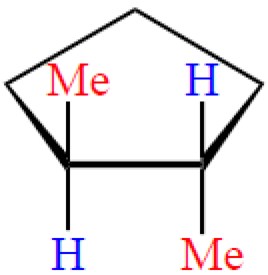

*trans*-1,2-Dimethylcyclopentane ($C_7H_{14}$)



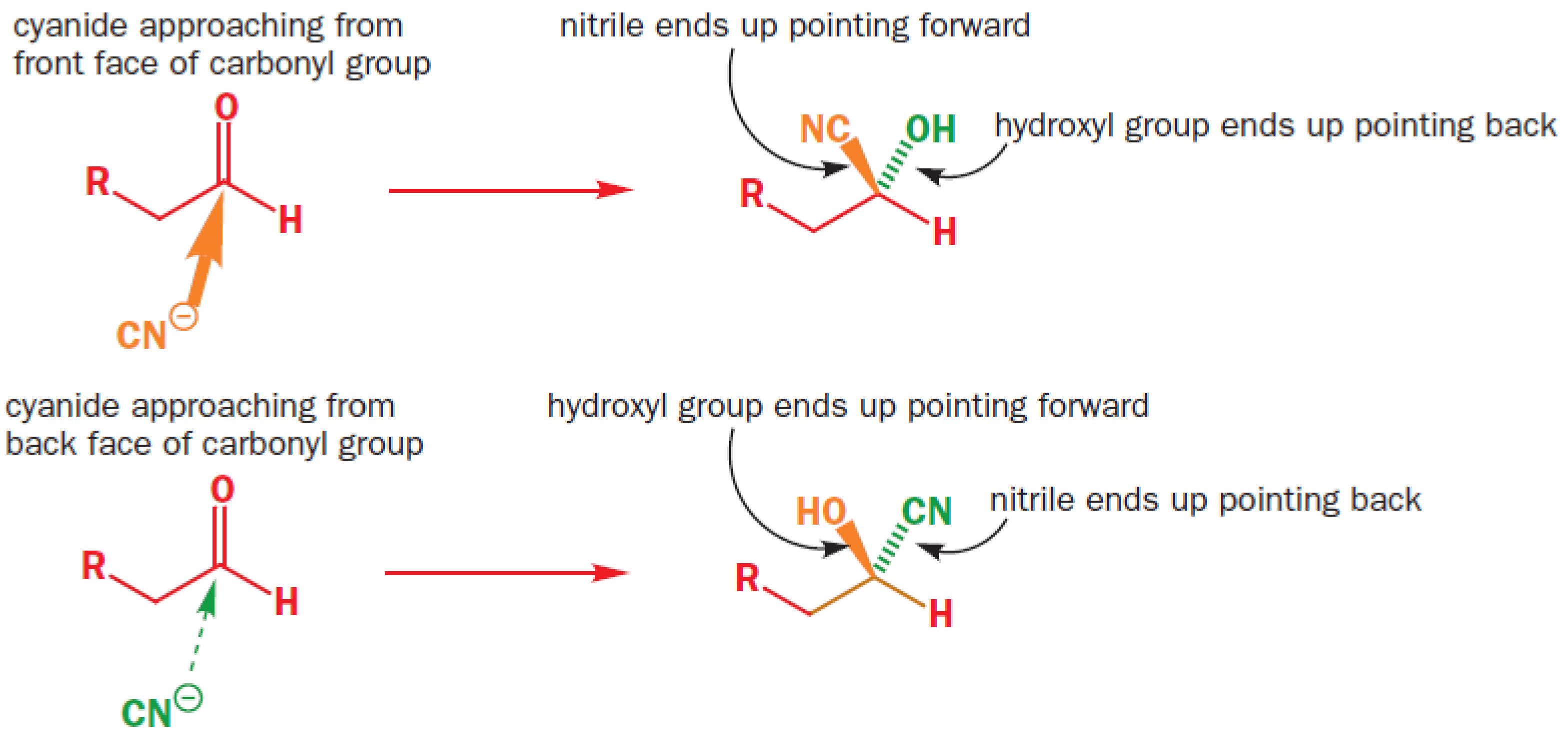

*trans*-1,2-Dimethylcyclopentane ($C_7H_{14}$)

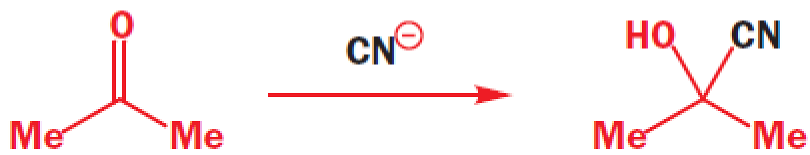


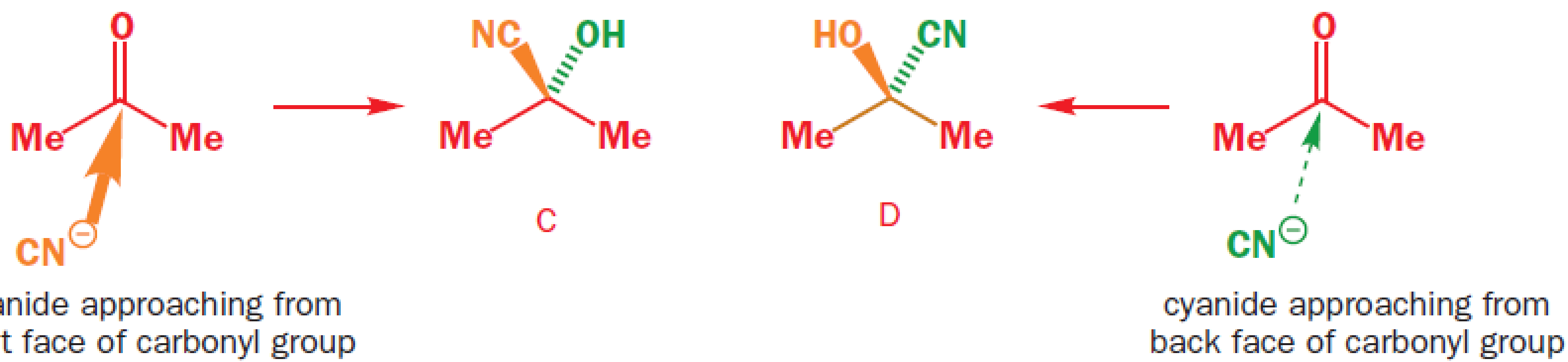

*cis*-1,2-Dimethylcyclopentane ($C_7H_{14}$)

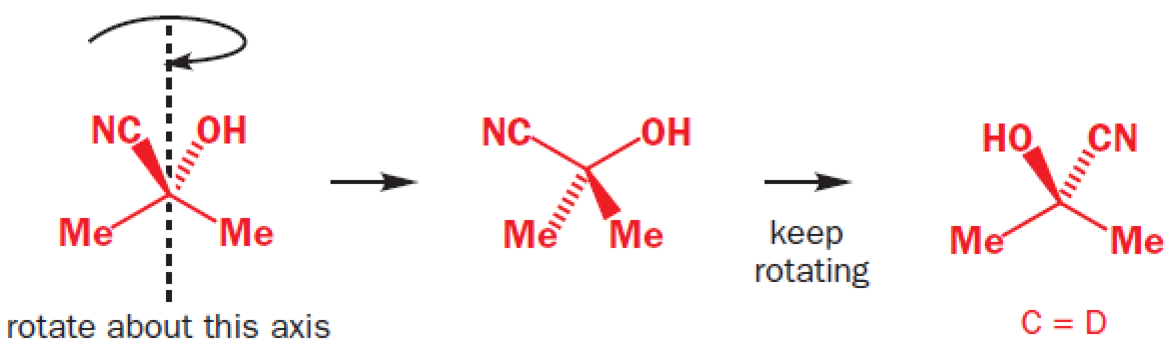


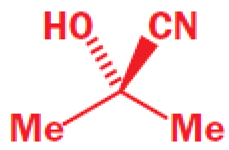

*trans*-1,2-Dimethylcyclopentane ($C_7H_{14}$)


cyanide approaching from front face of carbonyl group
O
R
H
CN
nitrile ends up pointing forward
NC
OH
hydroxyl group ends up pointing back
R
H
cyanide approaching from back face of carbonyl group
O
R
H
CN
hydroxyl group ends up pointing forward
HO
CN
nitrile ends up pointing back
R
H

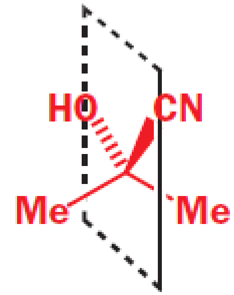

CN


cyanide approaching from front face of carbonyl group
C
D
cyanide approaching from back face of carbonyl group


rotate about this axis
keep rotating
C = D

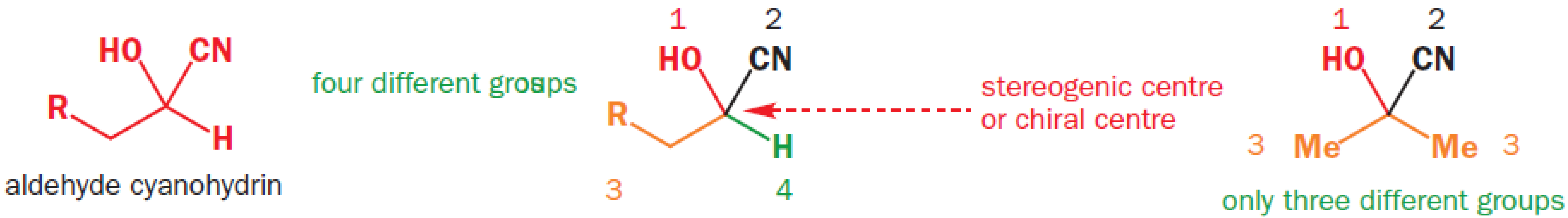



acetone cyanohydrin




plane of symmetry runs through central carbon, OH and CN

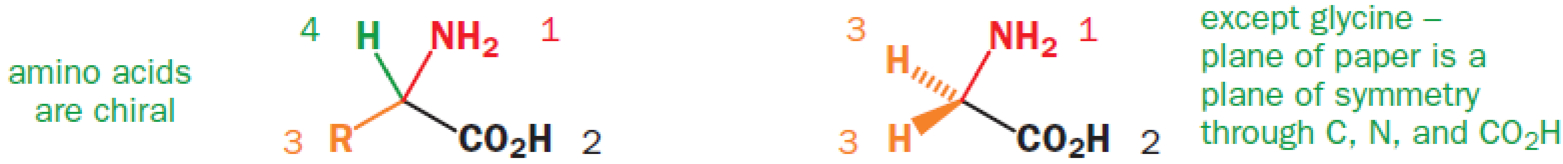
- Enantiomers are structures that are not identical, but are mirror images of each other.
- Structures are chiral if they cannot be superimposed upon their mirror image.
- Objects (and molecules) that are superposable on their mirror images are **achiral**.
- Any structure that has no plane of symmetry can exist as two mirror-image forms (**enantiomers**).
- Any structure with a plane of symmetry cannot exist as two enantiomers.




- If a molecule contains one carbon atom carrying four different groups it will not have a plane of symmetry and must therefore be chiral.
- A carbon atom carrying four different groups is a stereogenic or chiral centre.

- A racemic mixture is a mixture of two enantiomers in equal proportions.
- If the starting materials of a reaction are achiral, and the products are chiral, they will be formed as a racemic mixture of two enantiomers.

- All but one of the 20 amino acids that make up naturally occurring proteins are chiral, and all of them are classified as being left-handed (*S* configuration).

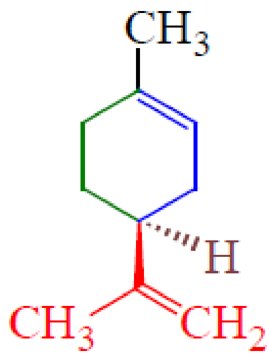



- The molecules of natural sugars are almost all classified as being right-handed (*R* configuration), including the sugar that occurs in DNA.

- DNA has a helical structure, and all naturally occurring DNA turns to the right.

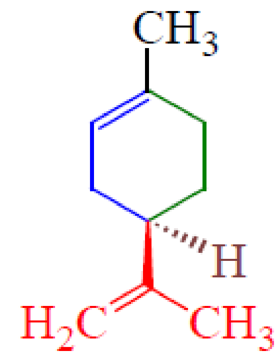
# Chirality & Biological Activity

**S**



lemon odor

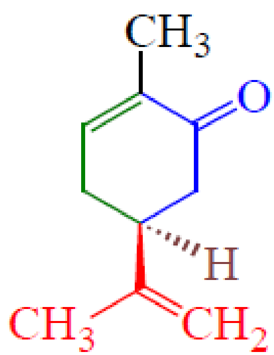
**Limonene**



orange odor

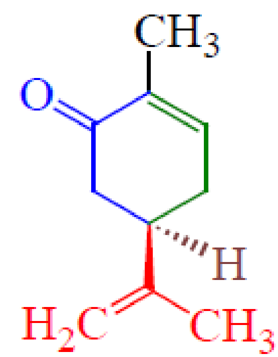
**R**

**S**



spearmint fragrance

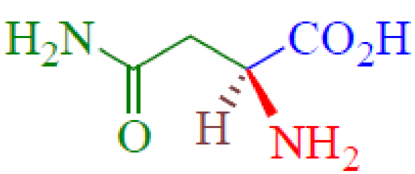
**Carvone**



caraway seed odor

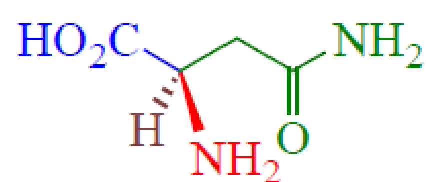
**R**

**S**



bitter taste

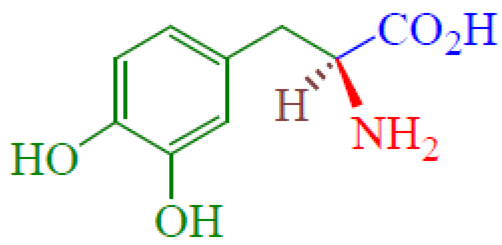
**Asparagine**



sweet taste

**R**

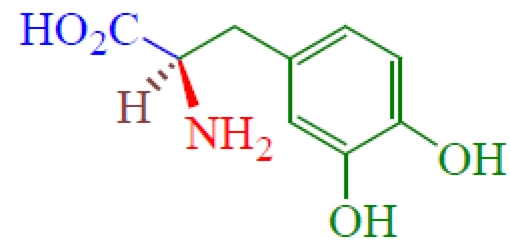
**S**



Anti-Parkinson's disease

**Dopa**

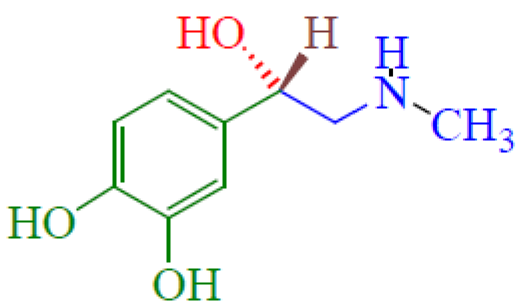
(3,4-dihydroxyphenylalanine)



**R**

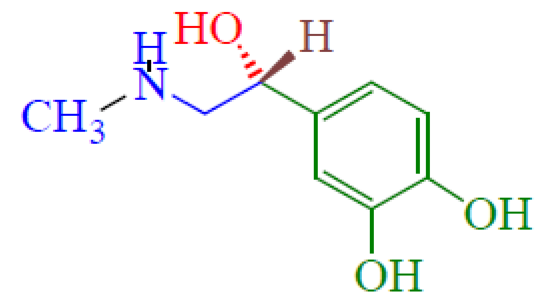
Toxic

**S**



Toxic

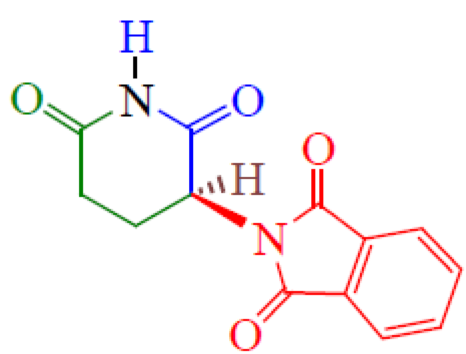
**Epinephrine**



**R**

hormone

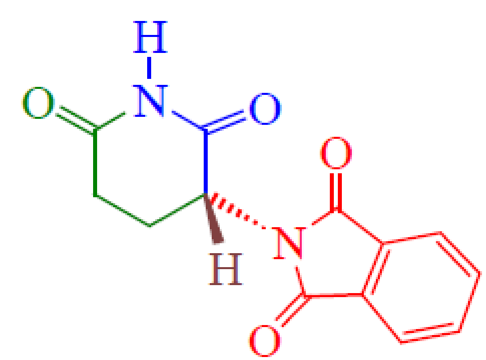
**S**



teratogenic activity

**Thalidomide**

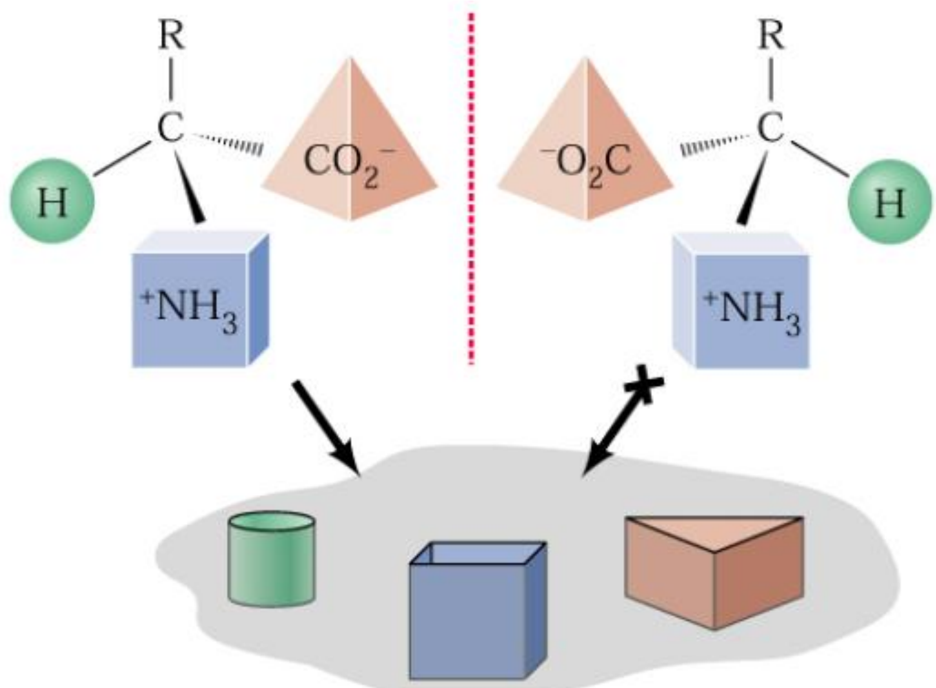
sedative, hypnotic



**R**

causes **NO** deformities

- The binding specificity for a chiral molecule (like a hand) at a chiral receptor site is only favorable in one way.
- If either the molecule or the biological receptor site had the wrong handedness, the natural physiological response will not occur.
- Because of the tetrahedral stereocenter of the amino acid, three-point binding can occur with proper alignment for only one of the two enantiomers.



# Tests For Chirality

1. Superposablility of the models of a molecule and its mirage:

- If the models are superposable, the molecule that they represent is **achiral**.
- If the models are non-superposable, the molecules that they represent are **chiral**.

2. The presence of a single tetrahedral stereocenter ⇒ chiral molecule.

3. The presence of a plane of symmetry ⇒ achiral molecule

- A plane of symmetry (also called a mirror plane) is an imaginary plane that bisects a molecule in such a way that the two halves of the molecule are mirror images of each other.
- The plane may pass through atoms, between atoms, or both.

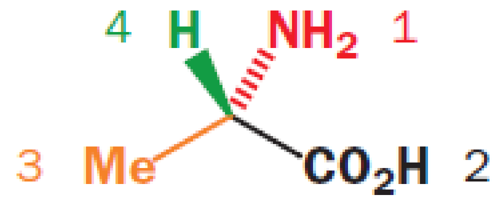
## Nomenclature of Enantiomers: The (*R*-*S*) System (Cahn-Ingold-Prelog System)

(*R*) and (*S*) are from the Latin words *rectus* and *sinister*:

- *R* configuration: clockwise (*rectus*, "**right**")
- *S* configuration: counterclockwise (*sinister*, "**left**")

1. **Each of the four groups attached to the stereocenter is assigned a priority.**

- Priority is first assigned on the basis of the **atomic number** of the atom that is directly attached to the stereocenter.
- The group with the **lowest atomic number** is given the **lowest priority**, **4**; the group with **next higher atomic number** is given the **next higher priority**, **3**; and so on.
- In the case of isotopes, the isotope of greatest atomic mass has highest priority.

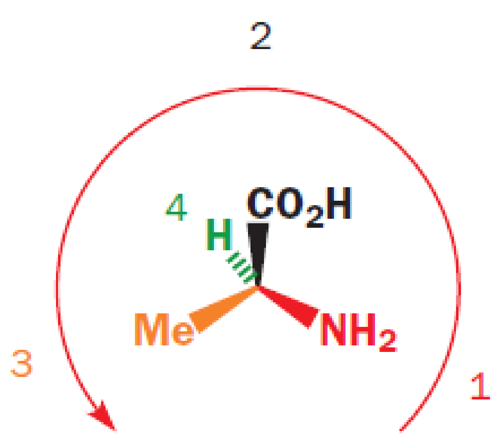


2. **Assign a priority at the first point of difference.**

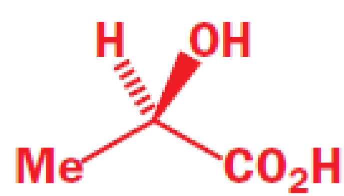
- When a priority cannot be assigned on the basis of the atomic number of the atoms that are directly attached to the stereocenter, then the next set of atoms in the unassigned groups are examined.

3. **View the molecule with the group of lowest priority pointing away from us.**

- If the direction from highest priority (4) to the next highest (3) to the next (2) is **clockwise**, the enantiomer is designated *R*.
- If the direction is **counterclockwise**, the enantiomer is designated *S*.

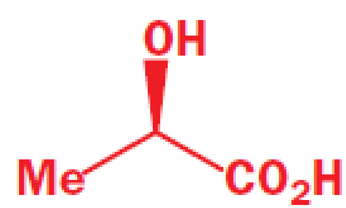



(*S*)-alanine



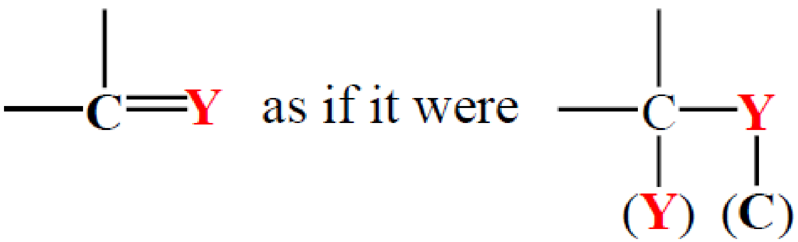

(*R*)-lactic acid

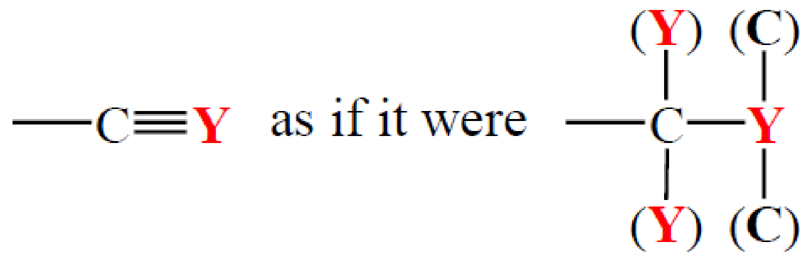
or

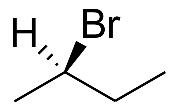



(*R*)-lactic acid

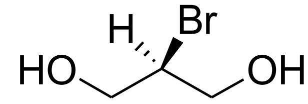
- The absolute stereochemistry of a **stereocenter**.
- **The sign of optical rotation is not related to the *R,S* designation**.
- Groups containing double or triple bonds are assigned priority as if both atoms were duplicated or triplicated.



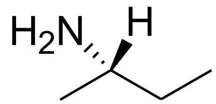





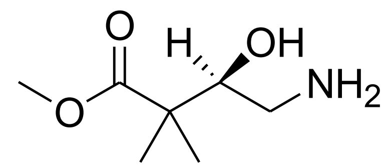
**1**



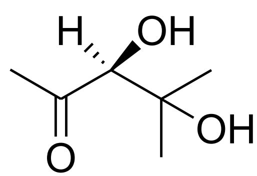
**2**



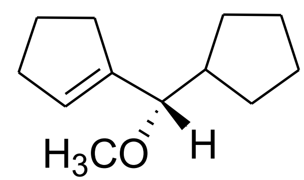
**3**



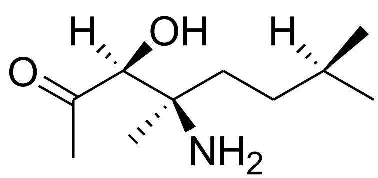
**4**



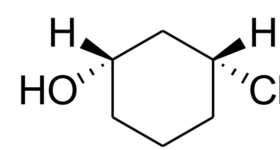
**5**



**6**



**7**



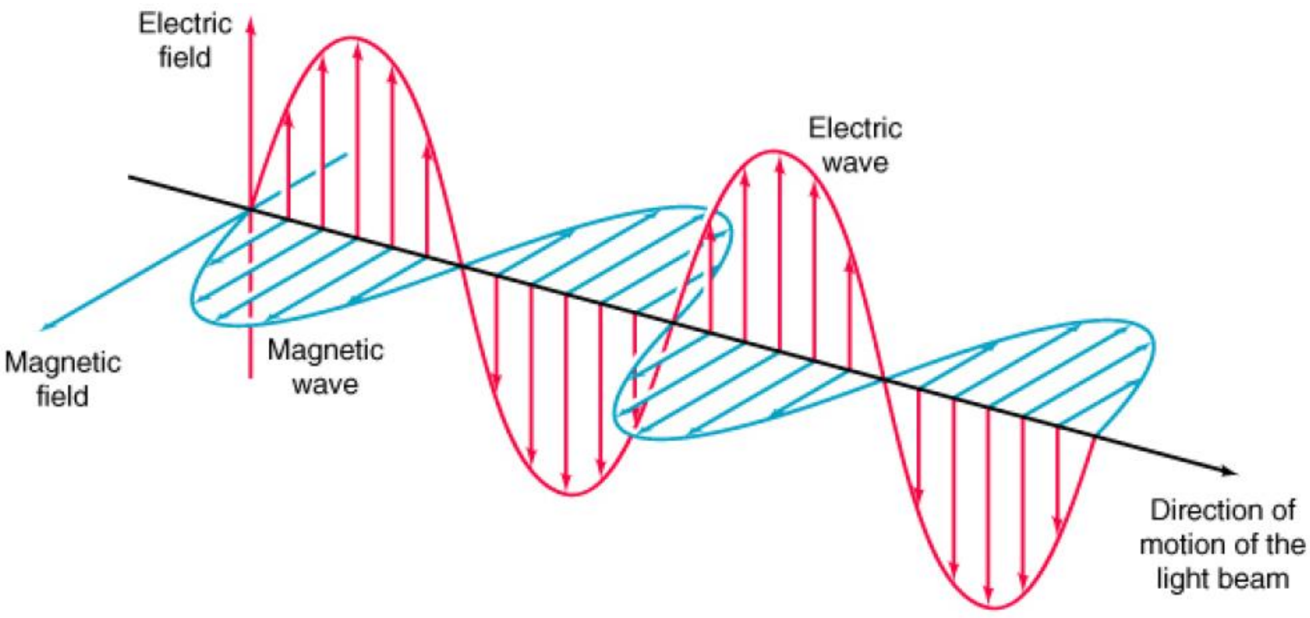
**8**

**Enantiomers** show **different behavior** only when they **interact** with other **chiral substances**.

- Enantiomers show **different rates of reaction** toward other **chiral** molecules.
- Enantiomers show **different solubilities** in **chiral** solvents that consist of a single enantiomer or an excess of a single enantiomer.

Enantiomers rotate the plane of **plane-polarized light** in equal amounts **but in opposite directions**.

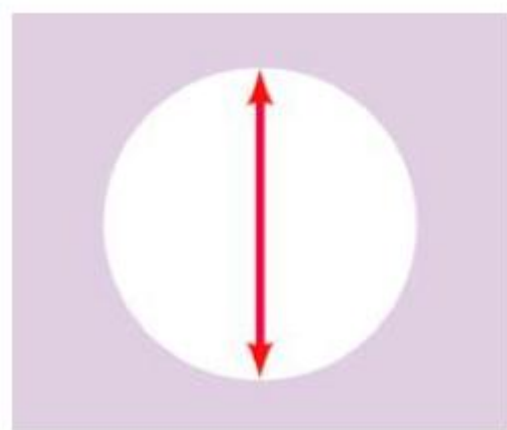
- Separate enantiomers are said to be **optically active** compounds.

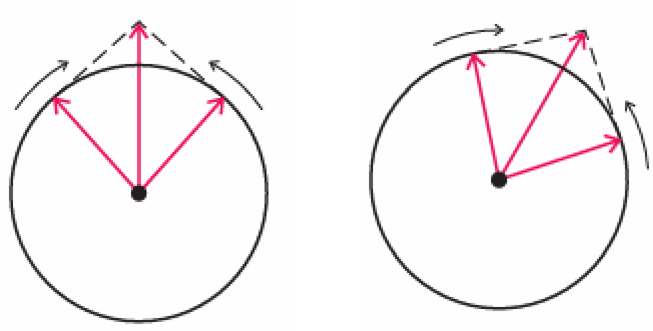


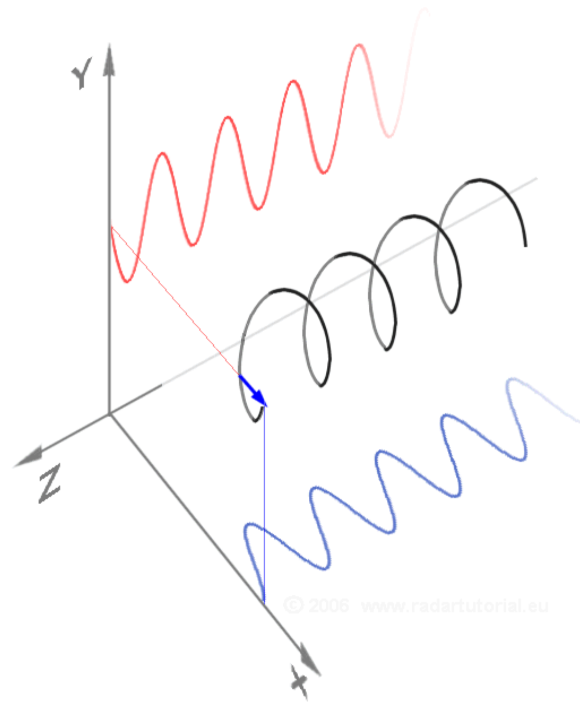

Oscillations of the **electric field** (and the **magnetic field**) are occurring in all possible planes **perpendicular** to the **direction of propagation**

# Plane Polarized Light

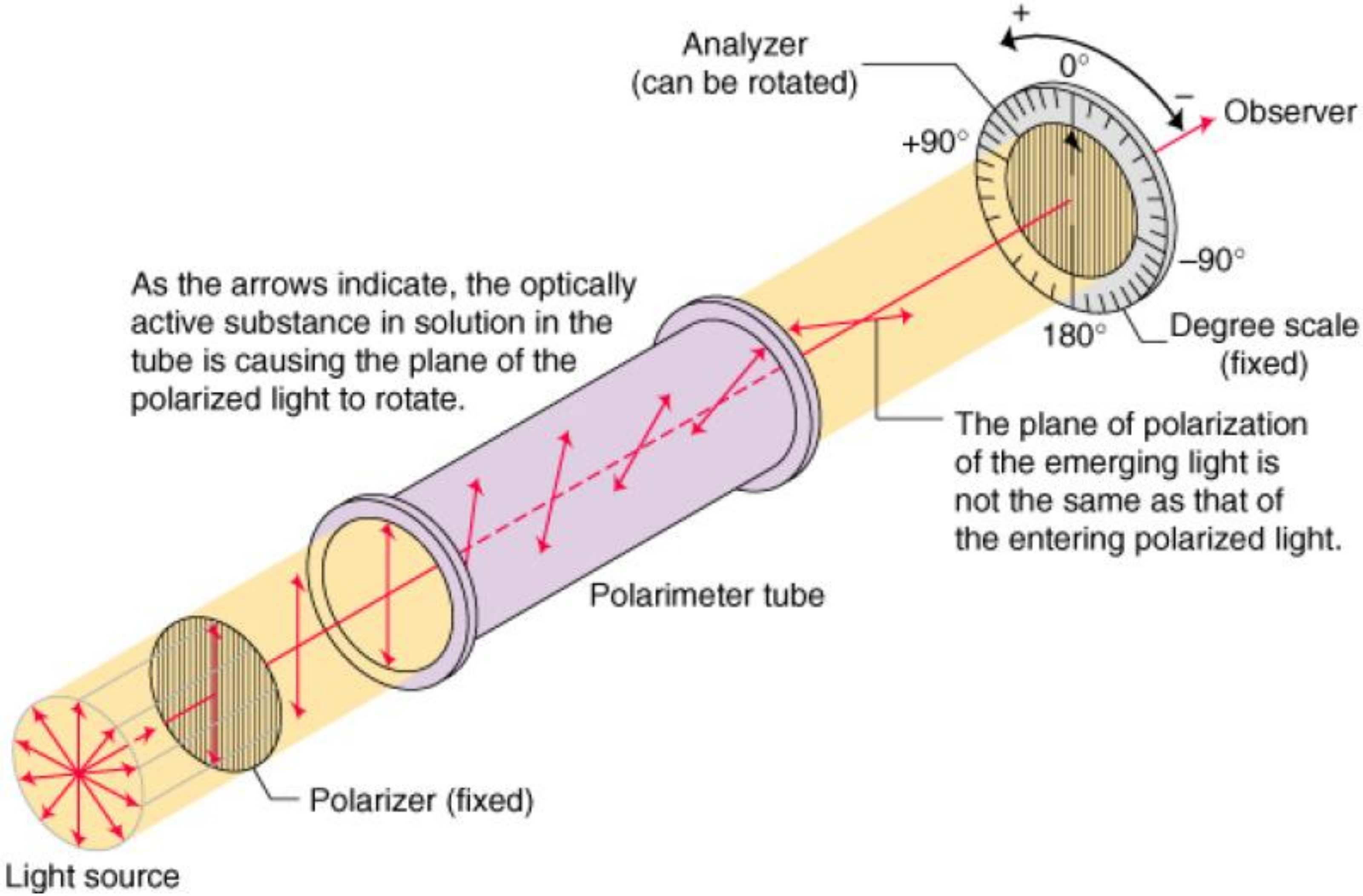
When ordinary light is passed through a **polarizer**, the polarizer interacts with the electric field so that the electric field of the light emerges from the polarizer (and the magnetic field perpendicular to it) is **oscillating only in one plane**

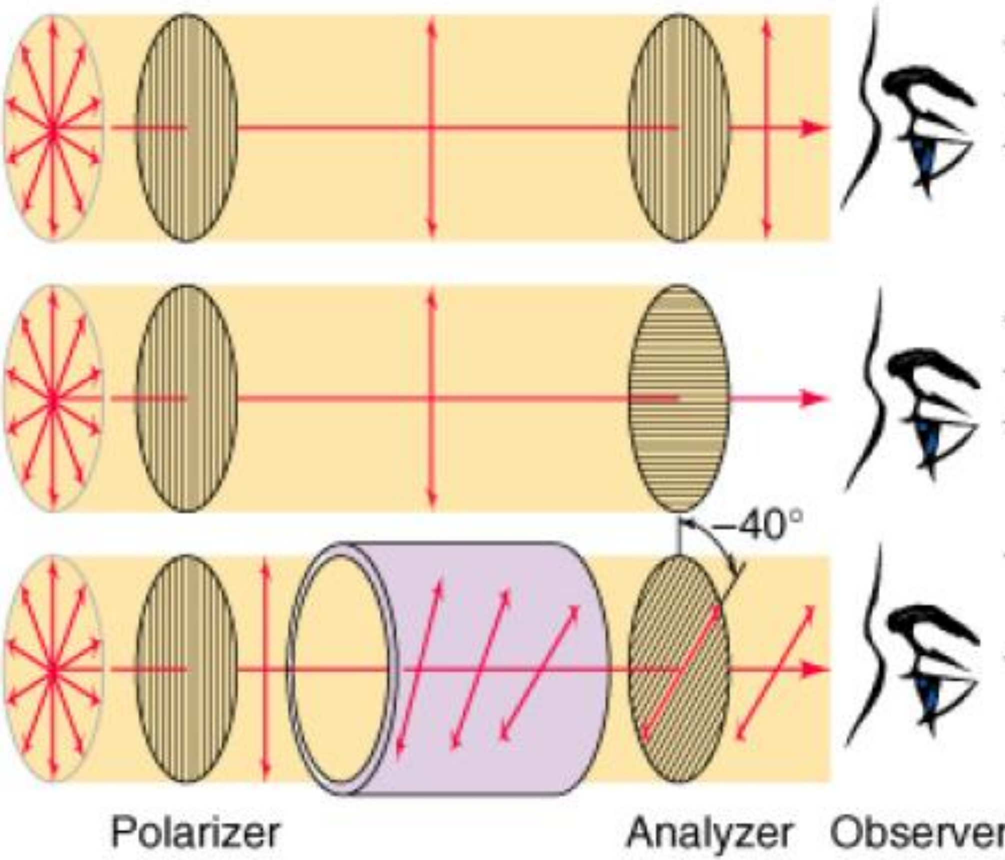






# Polarimeter





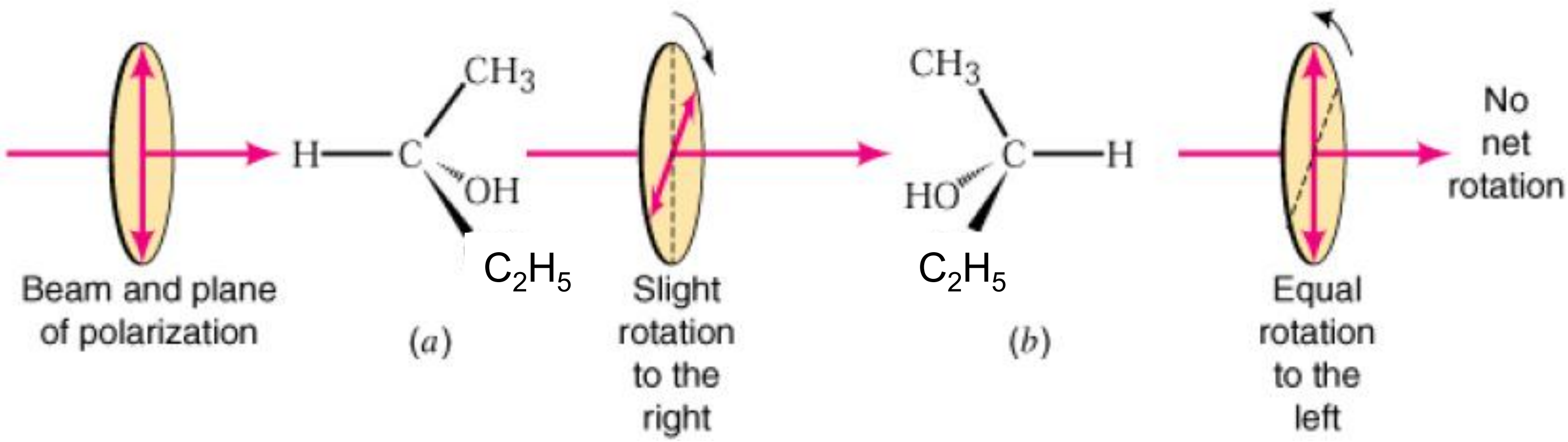

–Polarizer and analyzer are parallel.
–No optically active substance is present.
–Polarized light can get through analyzer.

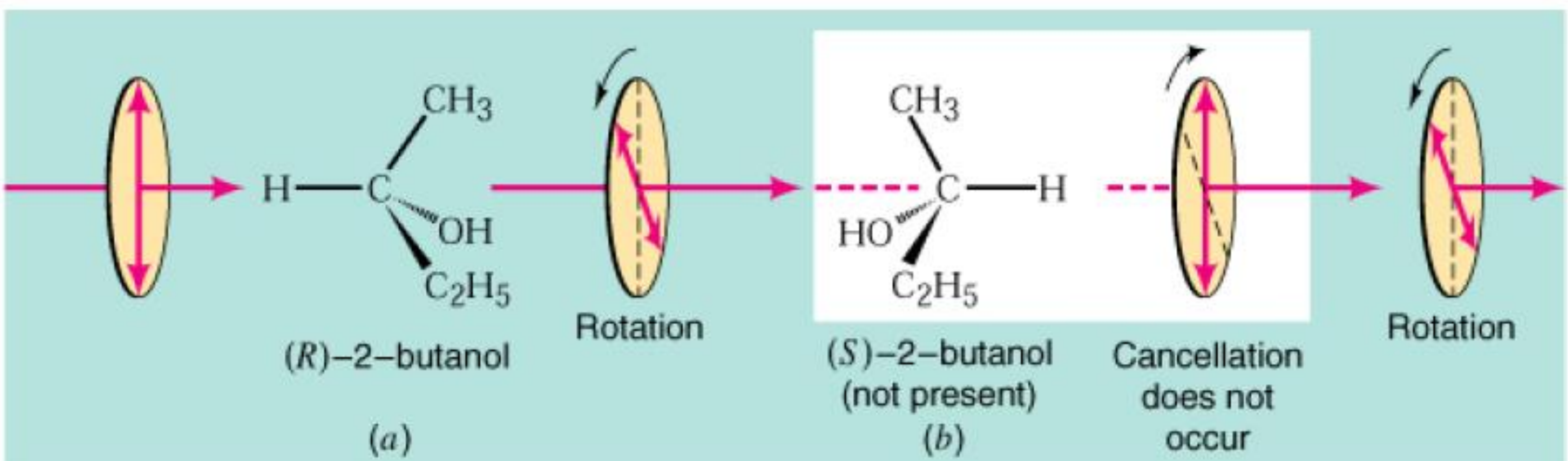
–Polarizer and analyzer are crossed.
–No optically active substance is present.
–No polarized light can emerge from analyzer.

–Substance between polarizer and analyzer is optically active.
–Analyzer has been rotated to the left (from observer's point of view) to permit rotated polarized light through (substance is levorotatory).

If the analyzer is rotated in a **clockwise** direction, the rotation, α (measured in degree) is said to be **positive** (+), and if the rotation is **counterclockwise**, the rotation is said to be **negative** (–).

A substance that rotates plane-polarized light in the **clockwise** direction is said to be **dextrorotatory**, and one that rotates plane-polarized light in a **counterclockwise** direction is said to be **levorotatory** (Latin: *dexter*, right; and *laevus*, left).


CH3
H—C—OH
C2H5
CH3
C—H
HO
C2H5
Beam and plane of polarization
(a)
Slight rotation to the right
(b)
Equal rotation to the left
No net rotation


CH3
H—C—OH
C2H5
(R)–2–butanol
(a)
Rotation
CH3
C—H
HO
C2H5
(S)–2–butanol (not present)
(b)
Cancellation does not occur
Rotation

**Specific rotation, [α]:**

$$[\alpha]_D^T = \frac{\alpha}{l \times c}$$

α: observed rotation

$l$: sample path length (dm)

$c$: sample concentration (g/mL)

$$[\alpha]_D^T = \frac{\text{Observed rotation, } \alpha}{\text{Path length, } l \text{ (dm) x Concentration of sample, } c \text{ (g/mL)}} = \frac{\alpha}{l \times c}$$

degrees $(g/mL)^{-1}$ $dm^{-1}$

- The specific rotation depends on the **temperature** and **wavelength** of light that is employed.
  - Na D-line: 589.6 nm = 5896 Å.
  - Temperature (*T*).
- The magnitude of rotation is dependent on the **solvent** when solutions are measured.

- The direction of rotation of plane-polarized light is often incorporated into the names of optically active compounds.

- No correlation exists between the configuration of enantiomers and the direction of optical rotation.

- No correlation exists between the (*R*) and (*S*) designation and the direction of optical rotation.

# Enantiomeric Excess

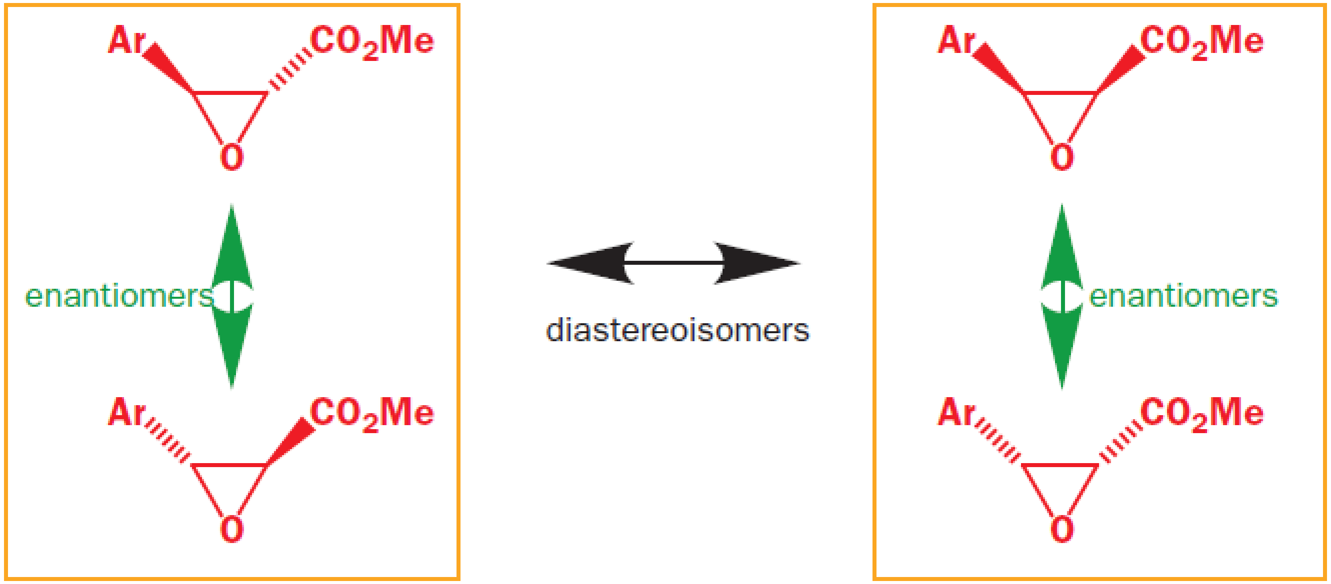
$$\% \text{ Enantiomeric excess } = \frac{M_+ - M_-}{M_+ + M_-} \text{ x } 100$$

Where M+ is the mole fraction of the dextrorotatory enantiomer, and $M_-$ the mole fraction of the levorotatory one.

$$\% \text{ optical purity } = \frac{[\alpha]_{\text{mixture}}}{[\alpha]_{\text{pure enantiomer}}} \text{ x } 100$$

- $[\alpha]_{\text{pure enantiomer}}$ value has to be available
- detection limit is relatively high (required large amount of sample for small rotation compounds)

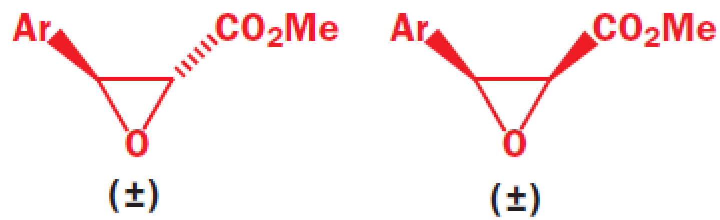
# Diastereoisomers




*trans* epoxide

*cis* epoxide

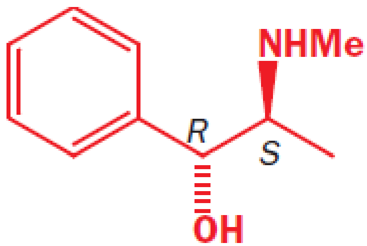
# Absolute and relative stereochemistry



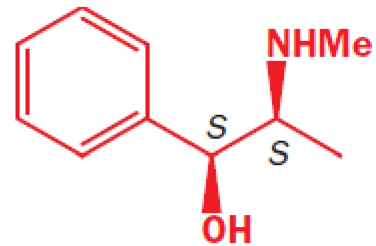

When the stereochemistry drawn on a molecule means ‘this diastereoisomer’, we say that we are representing **relative stereochemistry**; when it means ‘this enantiomer of this diastereoisomer’ we say we are representing its **absolute stereochemistry**. Relative stereochemistry tells us only how the stereogenic centres within a molecule relate to each other.

- A pair of enantiomers are mirror-image forms of the same compound and have opposite absolute stereochemistry.
- Two diastereoisomers are different compounds, and have different relative stereochemistry


- To go from one enantiomer to another, both stereogenic centres are inverted
- To go from one diastereoisomer to another, only one of the two is inverted.




(1*R*,2*S*)-(–)-ephedrine



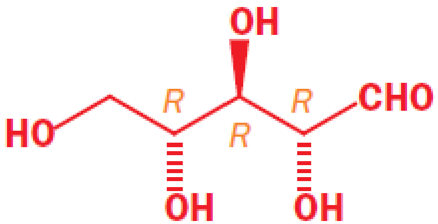

(1*S*,2*S*)-(+)-pseudoephedrine

| | **(1*R*,2*S*)-(–)-ephedrine** | **(1*S*,2*R*)-(+)-ephedrine** | **(1*S*,2*S*)-(+)-pseudoephedrine** | **(1*R*,2*R*)-(–)-pseudoephedrine** |
|---|---|---|---|---|
| m.p. | 40–40.5 °C | 40–40.5 °C | 117–118 °C | 117–118 °C |
| $[\alpha]_D^{20}$ | –6.3 | +6.3 | +52 | –52 |

The diastereoisomers are different compounds with different names and different properties, while the pair of enantiomers are the same compound and differ only in the direction in which they rotate polarized light.

Number of stereoisomers = $2^n$

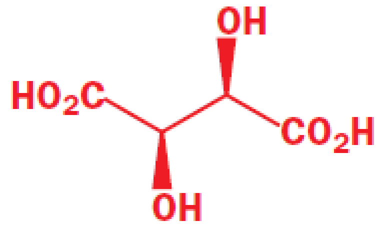
n = No. of stereogenic centres




(D)-ribose

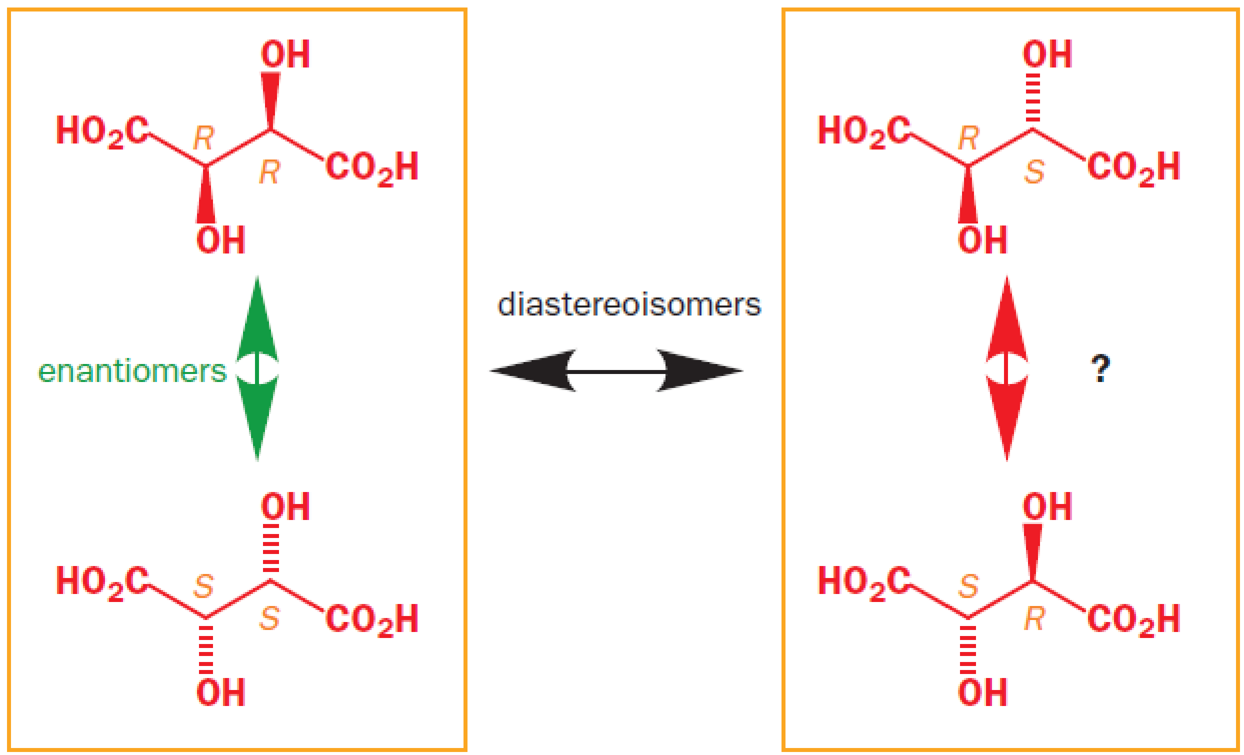
| | | | |
|---|---|---|---|
| *RRR* | *RRS* | *RSR* | *RSS* |
| *SSS* | *SSR* | *SRS* | *SRR* |

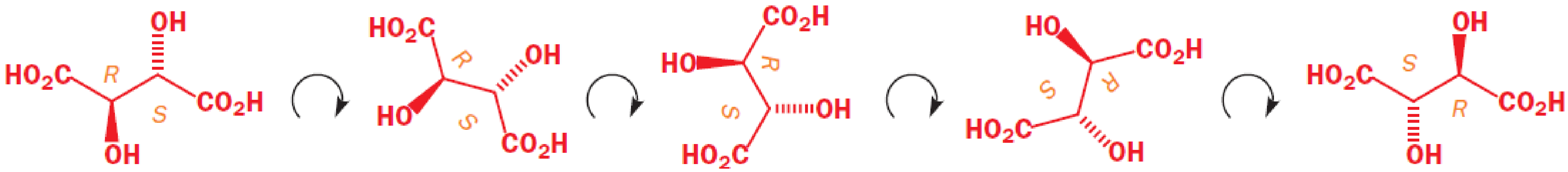
- A structure with n stereogenic centres can exist as $2^n$ stereoisomers.
- These stereoisomers consist of $2^{(n-1)}$ diastereoisomers, each of which has a pair of enantiomers (works only if all diastereoisomers are chiral)

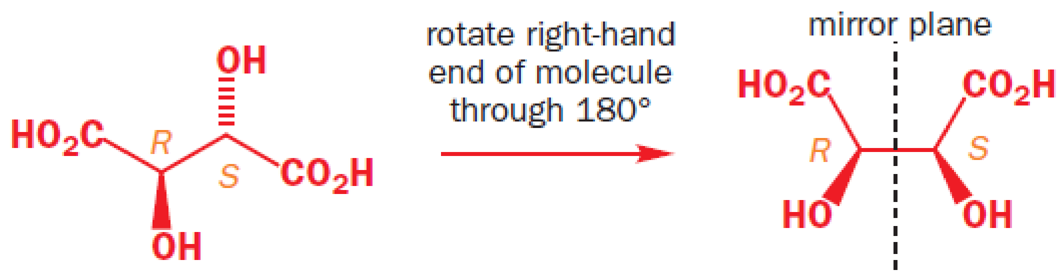



(+)-tartaric acid

OH groups *syn* | OH groups *anti*








| | Chiral diastereoisomer | | Achiral diastereoisomer |
|---|---|---|---|
| | (+)-tartaric acid | (–)-tartaric acid | *meso*-tartaric acid |
| $[\alpha]_D^{20}$ | +12 | –12 | 0 |
| m.p. | 168–170 °C | 168–170 °C | 146–148 °C |

Compounds that contain stereogenic centres but are themselves achiral are called *meso* compounds. This means that there is a plane of symmetry with *R* stereochemistry on one side and *S* stereochemistry on the other.